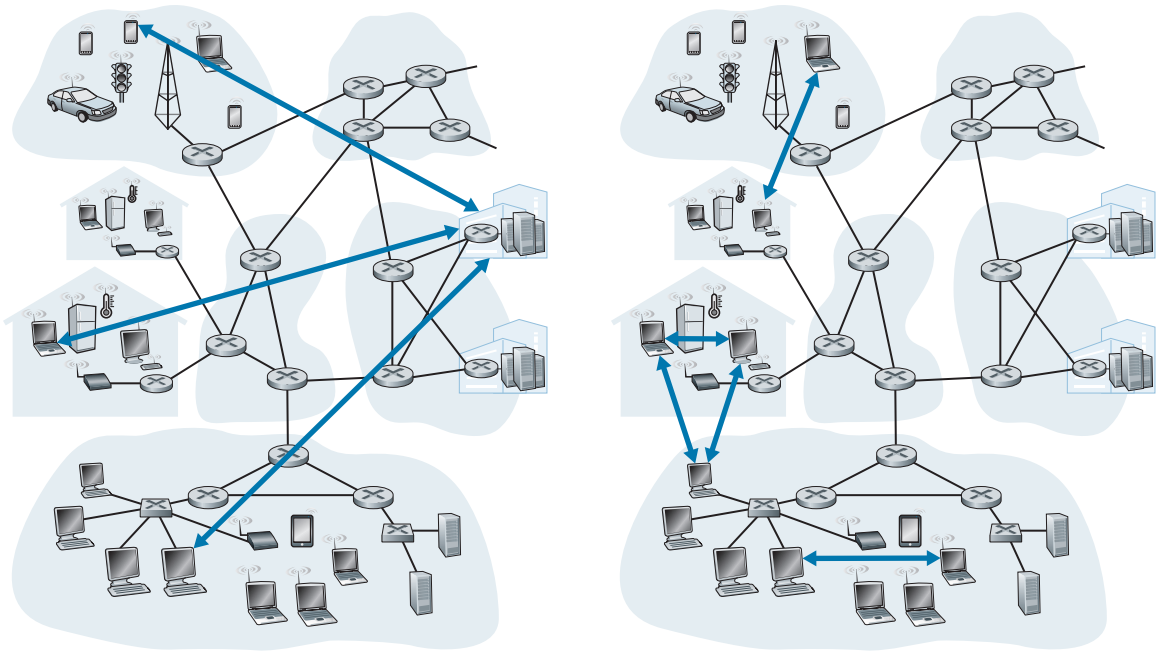

a. Client-server architecture

b. Peer-to-peer architecture

Figure 2.2 ♦ (a) Client-server architecture; (b) P2P architecture

without passing through a dedicated server, the architecture is called peer-to-peer. An example of a popular P2P application is the file-sharing application BitTorrent.

One of the most compelling features of P2P architectures is their **self-scalability**. For example, in a P2P file-sharing application, although each peer generates workload by requesting files, each peer also adds service capacity to the system by distributing files to other peers. P2P architectures are also cost effective, since they normally don't require significant server infrastructure and server bandwidth (in contrast with clients-server designs with datacenters). However, P2P applications face challenges of security, performance, and reliability due to their highly decentralized structure.

2.1.2 Processes Communicating

Before building your network application, you also need a basic understanding of how the programs, running in multiple end systems, communicate with each other. In the jargon of operating systems, it is not actually programs but **processes** that